Prime Numbers and the Zeta Function: A Short Guide

This guide introduces prime numbers, how they are found and counted, and the zeta function that connects them to the rest of mathematics and to physics.

1. What a prime number is

A prime number is a whole number greater than one whose only divisors are one and itself. Two, three, five, seven and eleven are prime. Four is not, because it is two times two, and nine is not, because it is three times three. A whole number greater than one that is not prime is called composite.

The number one is not counted as a prime. This is a choice rather than a discovery, but it is a useful one: if one were prime, a number could be written as a product of primes in many ways, by adding as many ones as we liked, and the most important fact about primes would need a long list of exceptions.

Two is the only even prime. Every other even number can be divided by two, so it has a divisor besides one and itself. This makes two an odd prime out in many statements, and mathematicians often speak of the odd primes when they want to leave it aside.

Primes have been studied for at least two and a half thousand years. Greek mathematicians knew how to find them, knew that there are infinitely many of them, and knew that every whole number is built from them in exactly one way. Much of what is still unknown about primes concerns how they are spread out among the other numbers.

2. Building every number from primes

The fundamental theorem of arithmetic says that every whole number greater than one can be written as a product of primes, and that this product is unique apart from the order of its factors. For example, 360 is two times two times two times three times three times five, and there is no other way to write it as a product of primes.

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360 = 2^3 x 3^2 x 5
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The existence part is easy: if a number is not prime, split it into two smaller factors, and keep splitting until every piece is prime. The uniqueness part is harder. It rests on a fact proved by Euclid: if a prime divides a product of two numbers, it must divide at least one of them.

Because of this theorem, primes are often called the atoms of arithmetic. Questions about whole numbers can often be turned into questions about their prime factors. The greatest common divisor of two numbers, for instance, is found by taking each prime to the smaller of the two powers in which it appears.

The theorem also explains why factoring large numbers matters in practice. Multiplying two large primes is quick, but recovering them from their product is believed to be very slow when the primes have hundreds of digits. Much of the encryption used on the internet depends on that difference.

3. There are infinitely many primes

Euclid gave a proof, more than two thousand years ago, that the primes never run out. Suppose there were only finitely many primes. Multiply all of them together and add one. The new number leaves a remainder of one when divided by any prime on the list, so none of them divides it. But every number greater than one has a prime factor, so there must be a prime missing from the list.

The proof does not say that the new number is itself prime. Two times three times five times seven times eleven times thirteen, plus one, is 30031, which is 59 times 509. What the proof shows is that some prime is missing from any finite list, and so no finite list can contain them all.

Many other proofs have been found since. One of the most important, due to Leonhard Euler in the eighteenth century, shows something stronger: the sum of the reciprocals of the primes grows without limit. This means the primes are not only infinite in number but fairly common, far more common than the perfect squares, whose reciprocals add up to a finite total.

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1/2 + 1/3 + 1/5 + 1/7 + 1/11 + ... grows without limit
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Euler's proof is the first step on the road that leads to the zeta function, which is described later in this guide.

4. Finding primes: the sieve of Eratosthenes

The oldest practical way to list primes is the sieve of Eratosthenes, named after a Greek scholar who lived in Alexandria around 240 BC. To find every prime up to a limit, write down all the numbers from two to the limit. Circle two, and cross out every larger multiple of two. Then circle the next number not crossed out, which is three, and cross out its multiples. Keep going.

The numbers circled are exactly the primes. It is enough to sieve with primes up to the square root of the limit: any composite number below the limit has a prime factor no larger than its square root, so it has already been crossed out by the time that point is reached.

The sieve is fast. Listing the primes below one million takes a modern computer a few milliseconds. Its weakness is memory, since every number up to the limit must be kept, so for very large ranges the sieve is run on one segment at a time.

To test a single large number, sieving is far too slow. Modern tests, such as the Miller-Rabin test, check a property that every prime has and that almost every composite number lacks. They can decide whether a number with hundreds of digits is prime in a fraction of a second.

5. How many primes there are

Counting primes shows a steady thinning. One in four of the numbers below one hundred is prime, but only about one in thirteen of those below one million. The function that counts the primes up to a number x is written pi(x), which has nothing to do with the circle constant.

The table below gives pi(x) for powers of ten, together with x divided by the natural logarithm of x, which is the estimate given by the prime number theorem.

x	Primes up to x	x / ln(x)	Ratio
10	4	4.3	0.921
100	25	21.7	1.151
1,000	168	144.8	1.161
10,000	1,229	1,085.7	1.132
100,000	9,592	8,685.9	1.104
1,000,000	78,498	72,382.4	1.084
10,000,000	664,579	620,420.7	1.071

The estimate is always a little low, but the ratio between the true count and the estimate gets closer to one as x grows. Carl Friedrich Gauss noticed this pattern as a teenager, around 1792, by studying tables of primes. It was proved a century later, in 1896, independently by Jacques Hadamard and Charles de la Vallee Poussin, using the zeta function.

6. The prime number theorem

The prime number theorem says that the number of primes up to x is roughly x divided by the natural logarithm of x, in the sense that the ratio of the two tends to one as x grows without limit.

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pi(x) ~ x / ln(x)
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Put another way, near a large number x, about one number in every ln(x) is prime. Near one million, ln(x) is about 13.8, so roughly one number in fourteen is prime. Near one trillion, ln(x) is about 27.6, and the primes are half as dense again.

A better estimate is the logarithmic integral, Li(x), which adds up one over the logarithm of every number up to x. Gauss suggested it, and it is remarkably accurate: for x equal to ten to the power of ten, Li(x) is off from the true count by only about three thousand, out of four hundred and fifty five million.

How large the error in Li(x) can be is one of the central questions of number theory. The Riemann hypothesis, if it is true, says that the error is never much larger than the square root of x. That is the best that could be hoped for, and it is what makes the hypothesis so important.

7. Euler and the zeta function

In 1737 Leonhard Euler studied the infinite sum of one over n to the power s, for numbers s greater than one. This sum is now called the zeta function, written zeta(s).

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zeta(s) = 1 + 1/2^s + 1/3^s + 1/4^s + ...
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Euler found that this sum over all whole numbers equals a product over all the primes. Each prime p contributes a factor of one divided by the quantity one minus one over p to the power s. This is called the Euler product.

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zeta(s) = product over primes p of 1 / (1 - 1/p^s)
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The Euler product is the fundamental theorem of arithmetic written as an equation: expanding each factor as a geometric series and multiplying them out gives every product of prime powers exactly once, which means every whole number exactly once. It links the primes, which are irregular, to a sum over all the whole numbers, which is perfectly regular.

Euler used it to prove again that there are infinitely many primes. As s comes down to one, the sum becomes the harmonic series, which grows without limit. If there were only finitely many primes, the product would stay finite, so there must be infinitely many.

8. The Basel problem

Before he found the Euler product, Euler made his name by solving the Basel problem, which had defeated the best mathematicians for ninety years. The problem asked for the exact value of the sum of the reciprocals of the squares.

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1 + 1/4 + 1/9 + 1/16 + ... = pi^2 / 6
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Euler announced in 1734 that the sum is pi squared over six, about 1.6449. In the language of the zeta function, zeta(2) equals pi squared over six. The appearance of pi, a number from geometry, in a sum about whole numbers was a great surprise.

Euler went on to find zeta(s) for every even number s. Each is a rational number times pi to the power s. For example, zeta(4) is pi to the fourth over ninety. The odd values are much more mysterious. In 1978 Roger Apery proved that zeta(3) is irrational, but nobody knows a simple formula for it, and very little is known about zeta(5), zeta(7) and the rest.

Combining Euler's two discoveries gives a pleasant fact: the chance that two whole numbers chosen at random have no common factor is six over pi squared, about 61 percent.

9. Riemann's paper of 1859

Bernhard Riemann, a German mathematician, wrote only one paper on number theory. Published in 1859 and only eight pages long, it changed the subject. Riemann treated zeta(s) as a function of a complex number s, a number with a real part and an imaginary part, rather than only a real one.

The sum that defines zeta(s) only makes sense when the real part of s is greater than one. Riemann showed how to extend the function to every complex number except s equal to one, where it has a pole. He also found a symmetry, the functional equation, that relates the value at s to the value at one minus s.

Riemann then gave an exact formula for counting primes. It expresses the number of primes up to x as a main term, close to Li(x), plus a correction for every zero of the zeta function. The zeros act like the notes of a chord: each adds a wave to the count, and together they reproduce the exact step where each prime appears.

The zeta function has trivial zeros at the negative even numbers, minus two, minus four and so on. All of its other zeros, the nontrivial ones, lie in the critical strip, where the real part of s is between zero and one. Where exactly they lie decides how well the primes follow the prime number theorem.

10. The Riemann hypothesis

Riemann suggested, almost in passing, that all the nontrivial zeros of the zeta function have real part exactly one half. They would then lie on a single line in the middle of the critical strip, now called the critical line. This suggestion is the Riemann hypothesis.

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every nontrivial zero of zeta(s) has real part 1/2
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The hypothesis has been checked for the first ten trillion zeros, and every one of them lies on the critical line. It has not been proved. In 1900 David Hilbert put it on his list of the most important open problems, and in 2000 the Clay Mathematics Institute made it one of its seven Millennium Prize Problems, with a prize of one million dollars for a proof.

If it is true, the primes are as regular as they could possibly be: the error in the prime number theorem is about the square root of x, no larger. Hundreds of theorems have been proved on the assumption that it is true, so a proof would settle all of them at once. A disproof would be just as dramatic.

Some partial results are known. Godfrey Hardy proved in 1914 that infinitely many zeros lie on the critical line, and later work showed that at least two fifths of them do. None of these comes close to all of them.

11. Primes and the zeta function in physics

In 1972 the mathematician Hugh Montgomery was studying the spacing between zeros of the zeta function. Over tea at Princeton, the physicist Freeman Dyson recognized his formula: it was the same as the spacing between energy levels of heavy atomic nuclei, as described by random matrix theory.

Since then, computations by Andrew Odlyzko of millions of zeros high on the critical line have matched the predictions of random matrix theory with great accuracy. Many researchers now suspect that the zeros are the energy levels of some quantum system that has not yet been found. If it were found, the Riemann hypothesis might follow from the laws of quantum mechanics.

The zeta function appears elsewhere in physics as well. The value zeta(3) turns up in the energy of black body radiation and in quantum electrodynamics. The Casimir effect, a tiny force between two metal plates in a vacuum, is often explained with a method called zeta function regularization, which gives a finite meaning to sums that would otherwise grow without limit.

These links are still not fully understood, which is part of what makes the subject so active.

12. Open questions about primes

Besides the Riemann hypothesis, many simple questions about primes are still open. The twin prime conjecture says that there are infinitely many pairs of primes that differ by two, such as 11 and 13, or 41 and 43. In 2013 Yitang Zhang proved that there are infinitely many pairs of primes that differ by less than seventy million, and later work brought that bound down to 246.

Goldbach's conjecture, from a letter Christian Goldbach wrote to Euler in 1742, says that every even number greater than two is the sum of two primes. It has been checked by computer up to four quintillion, and in 2013 Harald Helfgott proved the weaker version for odd numbers: every odd number greater than five is the sum of three primes.

Other questions ask whether there are infinitely many primes of the form n squared plus one, or how long a run of composite numbers can be found below a given size. Each sounds simple, and each has resisted the best efforts of mathematicians for a century or more.

Prime numbers are among the oldest objects of study in mathematics, and they still hold some of its deepest mysteries.

13. Primes in arithmetic progressions

Apart from two and five, every prime ends in the digit one, three, seven or nine. A natural question is whether the primes are shared out evenly among these four endings. They are: in the long run, a quarter of the primes end in each digit.

This is a special case of a theorem proved by Peter Gustav Lejeune Dirichlet in 1837. Take any two whole numbers a and q with no common factor. Then the progression a, a plus q, a plus two q, and so on contains infinitely many primes. For example, there are infinitely many primes that leave a remainder of one when divided by four, such as 5, 13, 17 and 29, and infinitely many that leave a remainder of three, such as 3, 7, 11 and 19.

Dirichlet's proof was a landmark because it used analysis, the mathematics of limits and continuous functions, to answer a question about whole numbers. He built functions much like the zeta function, now called Dirichlet L-functions, with one for each way of sorting numbers by their remainders.

Although the primes are shared out evenly in the long run, there are curious biases along the way. Primes that leave a remainder of three when divided by four are ahead of those that leave a remainder of one for most of the way. This is known as Chebyshev's bias, after Pafnuty Chebyshev, who noticed it in 1853.

In 2016 Robert Lemke Oliver and Kannan Soundararajan found a related surprise: consecutive primes seem to avoid repeating their last digit. A prime ending in one is followed by another prime ending in one less often than chance would suggest.

14. Mersenne primes and perfect numbers

A Mersenne number is one less than a power of two, such as 3, 7, 31 and 127. They are named after Marin Mersenne, a French monk who studied them in the seventeenth century. A Mersenne number can only be prime if its exponent is prime, but a prime exponent is not enough: two to the eleventh minus one is 2047, which is 23 times 89.

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M(p) = 2^p - 1
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Mersenne primes are linked to perfect numbers, numbers equal to the sum of their proper divisors. Six is perfect, because one plus two plus three is six, and so is 28. Euclid showed that whenever two to the p minus one is prime, multiplying it by two to the p minus one gives a perfect number. Euler later proved that every even perfect number arises this way. Whether an odd perfect number exists is still unknown.

The largest known primes are almost all Mersenne primes, because there is a fast test for them, the Lucas-Lehmer test. The Great Internet Mersenne Prime Search, a project in which volunteers lend their computers, has found every record prime since 1996. The largest known prime, found in 2024, has more than forty one million digits.

Only about fifty Mersenne primes are known. It is believed that there are infinitely many of them, but this has not been proved.

15. Primes and cryptography

Prime numbers protect much of the information sent over the internet. The RSA system, published in 1977 by Ron Rivest, Adi Shamir and Leonard Adleman, uses two large secret primes. Their product is made public and is used to lock messages; only someone who knows the two primes can unlock them.

The security of RSA rests on the difficulty of factoring. Multiplying two primes of three hundred digits each takes a computer a tiny fraction of a second, but no known method running on ordinary computers can recover the primes from their product in any reasonable time.

Making RSA keys needs a supply of large primes. The prime number theorem says they are plentiful: about one in every seven hundred numbers with three hundred digits is prime, so a computer picks random numbers of the right size and tests them until it finds one. Fast primality tests make this quick.

Other systems use primes in different ways. Elliptic curve cryptography works with points on curves whose coordinates are taken modulo a large prime, and gives the same security as RSA with much shorter keys.

A large enough quantum computer could factor large numbers quickly, using an algorithm found by Peter Shor in 1994. For this reason, new systems based on other hard problems are now being adopted, in what is called post-quantum cryptography.

16. Gaps between primes

The gap between two consecutive primes is the difference between them. Apart from the gap of one between two and three, every gap is even. On average, the gap near a number x is about ln(x), as the prime number theorem suggests, but single gaps can be much smaller or much larger.

Long gaps are easy to build. The numbers from one hundred factorial plus two to one hundred factorial plus one hundred are all composite, because the first is divisible by two, the next by three, and so on. So there are gaps as long as we like, although they appear only among very large numbers.

Short gaps are much harder. The smallest possible gap between odd primes is two, and the twin prime conjecture says that this gap occurs infinitely often. Zhang's result of 2013, improved by James Maynard, Terence Tao and the Polymath project, shows that some gap of at most 246 occurs infinitely often.

There are also limits on how long gaps can be. Bertrand's postulate, proved by Chebyshev in 1852, says that there is always a prime between any number n and two n. It is believed that there is always a prime between any two consecutive squares, but this is Legendre's conjecture and is still open.

17. Computing the zeros of the zeta function

Riemann himself computed the first few zeros of the zeta function by hand, as notes found in his papers decades later revealed. The first zero lies on the critical line at a height of about 14.13.

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first zeros: 1/2 + 14.13i, 1/2 + 21.02i, 1/2 + 25.01i
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In 1903 Jorgen Gram checked the first fifteen zeros. Alan Turing, better known for his work on computing and code breaking, designed a method in the 1950s for checking that no zeros are missed, and used one of the first electronic computers to check more than a thousand of them.

Today, a method by Andrew Odlyzko and Arnold Schonhage makes it possible to compute many zeros at once. The first ten trillion zeros have all been found on the critical line, and zeros near the ten to the twenty second zero have been computed to study their spacing.

No amount of computing can prove the Riemann hypothesis, since there are infinitely many zeros. But the computations have found no exception, and they have revealed the statistical patterns that link the zeros to random matrices and physics.

18. A short timeline

Around 300 BC, Euclid proves that there are infinitely many primes and lays the groundwork for the fundamental theorem of arithmetic. Around 240 BC, Eratosthenes describes his sieve.

In 1734 Euler solves the Basel problem, and in 1737 he finds the Euler product. In 1742 Goldbach writes to Euler with his conjecture. Around 1792 Gauss guesses the prime number theorem from tables of primes.

In 1837 Dirichlet proves his theorem on primes in arithmetic progressions. In 1859 Riemann publishes his paper on the zeta function and states his hypothesis. In 1896 Hadamard and de la Vallee Poussin prove the prime number theorem.

In 1914 Hardy proves that infinitely many zeros lie on the critical line. In 1972 Montgomery and Dyson find the link between the zeros and random matrices. In 1977 RSA is published. In 2013 Zhang proves that bounded gaps between primes occur infinitely often, and Helfgott proves the weak Goldbach conjecture.

The Riemann hypothesis, the twin prime conjecture and Goldbach's conjecture all remain open.